Integer-Only LLM Training via VFR Shell Arithmetic

Neural Network Weight Updates Through Exact Integer Remainder Accumulation on Harmonic Base-32 Octaves

Registry: [CKS-MATH-134-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-10-2026] → [CKS-MATH-104-2026] → [CKS-MATH-133-2026] → [CKS-MATH-134-2026]

Parent Framework: [CKS-0-2026]

DOI: 10.5281/zenodo.18959969

Date: March 11, 2026

Domain: Machine Learning / Integer Arithmetic / Neural Network Training

Status: CKS has been invalidated. The math does not compile, all papers in the series are falsified. Next steps: [CKS-NEXT-1-2026]

Old Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Classification: Theory of Everything from First Principles

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Lexicon: [CKS-LEX-12-2026]

ABSTRACT

We demonstrate neural network training using exclusively integer arithmetic, eliminating all floating-point computation from the forward pass, backward pass, and weight update. Weights are represented as VFR (Value-Factor-Remainder) tuples [V, F, R] where V is an i32 integer shell value, F is an implicit per-layer harmonic octave factor (power of 32), and R is an i16 remainder accumulating gradient pressure. Weight updates occur through shell transitions: gradients accumulate in R via integer addition; when $|R| \geq 32$ (one harmonic octave), V increments or decrements and R resets via modulo. No weight change occurs between transitions — the shell is stable until evidence accumulates. We

prove: (1) **Integer forward pass —** matrix multiplication via i32 multiply-accumulate with i64 accumulators and bit-shift normalization produces exact activations, (2) **Integer backward pass —** chain rule over integer operations produces exact integer gradients, (3) **Shell convergence —** gradient accumulation in R preserves signals below the precision floor that floating-point arithmetic destroys, (4) **Deterministic training —** identical input always produces identical output because integer arithmetic is associative and commutative, (5) **Harmonic octave system —** base-32 counting provides a universal scale from Planck length to observable universe in 65 octaves with all scale conversions as bit shifts, (6) **Structural interpretability —** weight nesting depth reveals information density without probing or ablation, (7) **Commodity hardware —** training runs on CPU with 64 GB RAM, no GPU required. Working implementation in Zig with zero external dependencies demonstrates the complete pipeline: tokenize, train, infer, evaluate. Loss decreases. Shells transition. Output tokens emerge from vocabulary learned through integer-only gradient descent.

Central claim: Floating-point arithmetic is not required for neural network training. Integer shell mechanics with exact remainder tracking provide a complete, deterministic, interpretable alternative that preserves gradient signals floats destroy and enables verification impossible in $\mathbb{R}$.

I. THE NUMBER 19

A weight sits at shell 0. During training, gradients arrive.

Step 1: gradient +7 → R = 7. No change.

Step 2: gradient +5 → R = 12. No change.

Step 3: gradient +4 → R = 16. No change.

Step 4: gradient +3 → R = 19. No change.

Step 5: gradient +8 → R = 27. No change.

Step 6: gradient +6 → R = 33. Shell transition.

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33 mod 32 = 1. V increments. R resets to 1.
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The number 19 is not a value. It is pressure. Nineteen thirty-seconds of the evidence needed to move this weight to the next stable state. Exact. Tracked. Sitting in the remainder field, waiting.

In floating-point training, each of those gradients — 7, 5, 4, 3, 8, 6 — might individually fall below the precision floor of BF16 or FP32 and vanish. The optimization signal existed but the number format destroyed it. The weight might never move. The model might never learn the pattern those gradients encode.

In integer shell training, every gradient contribution is preserved as an integer added to R. The weight moves exactly when 32 units of directional evidence have accumulated. Not before. Not after. Not approximately. Exactly when.

If the evidence reverses — negative gradients arrive after R = 19 — the pressure retreats: 19, 15, 9, 3. The weight does not move. The system tracks net pressure in both directions with perfect fidelity. No information is destroyed at any step.

This is the entire proposal in one example. Replace floating-point weights with integer shells. Track gradient pressure as exact integer remainders. Let weights transition between shells only when the accumulated evidence demands it.

II. THE PROBLEM: FLOAT INFORMATION LOSS

2.1 Three Values Compressed to One

A number has three natural components:

Value (V): the numerator — what you are counting.

Factor (F): the denominator — what scale you are counting at.

Remainder (R): the tracked residual — what pressure exists toward the next state.

The history of numerical computing compressed [V, F, R] → [V, F] → [V] → a single float. Each step discarded structural information. Floating point is the final stage: one number encoding what three integers describe exactly.

The consequence: the entire field of numerical analysis — epsilon tolerances, condition numbers, compensated summation, Kahan accumulators, re-orthogonalization, mixed-precision training — exists to recover information discarded when F and R were dropped.

2.2 The Base Mismatch

Binary floats represent values as sums of powers of 2. The statistical structure of language — token co-occurrence probabilities, syntactic patterns, semantic relationships — has no natural alignment to powers of 2.

The fraction 1/3 is unwritable in binary. The fraction 1/7 is unwritable. The fraction 6/5 = 1.2 is unwritable — it becomes 1.001100110011… repeating forever. Every time a model computes or stores such a value, it rounds. Every forward pass, every backward pass, every weight update introduces rounding that cannot be recovered.

In VFR notation: 1/3 = [1, 3, 0]. 1/7 = [1, 7, 0]. 6/5 = [6, 5, 0]. Three integers. Exact. Finite. Supporting binary equality.

2.3 What Gets Lost in LLM Training

Gradient swamping. When a weight has magnitude 1.0 in BF16 and a gradient update has magnitude 1e-7, the update vanishes. The precision gap between adjacent representable values is larger than the update. The optimization signal existed — the model computed it — but the number format destroyed it. In VFR shell training, that gradient adds to R exactly and accumulates until it triggers a shell transition.

Soft confusion between similar patterns. When two distinct concepts are encoded as weight vectors differing only in low-order bits, the attention mechanism may not reliably distinguish them. The dot-product similarity scores for the correct and incorrect patterns may differ by less than the float rounding error. VFR attention scores are exact integer ratios — the distinction is preserved through every layer.

Equality failure. Floating-point arithmetic does not support equality. After a sequence of operations, two values that should be identical are merely close. The entire epsilon-tolerance infrastructure of numerical computing is a consequence of this single property. VFR integer arithmetic supports binary equality. Values are equal or they are not.

III. THE HARMONIC OCTAVE SYSTEM

3.1 Base 32^{-1}

The fundamental counting unit is 32^{-1}. Since $32 = 2^5$, every scale operation is a 5-bit shift in binary hardware.

This is a harmonic system. Musical octaves double frequency ($\times 2$ per step). This system multiplies by 32 per step ($\times 2^5$). Five doublings per octave — a harmonic ladder where each rung is a 5-bit doubling matching the exponential scaling by which physical systems naturally organize.

Every value is a VFR tuple: [V, octave, R].

V is an integer value (i32). Octave is which power of 32 (u8). R is the remainder, modulo 32 (i16). The octave field replaces the denominator. Multiplying or dividing by F is a bit shift by 5 $\times$ octave. No division hardware needed. No floating-point logic. Shift.

3.2 The Universal Scale

The notation describes everything from the Planck length to the observable universe:

Octave 0: Planck length (~1.6×10^{-35} m)

Octave 2: Planck particle scale

Octave 10: Nuclear scale (~10^{-14} m)

Octave 22: Human-perceptible (~10^{-3} m)

Octave 37: Human heart

Octave 40: Human body

Octave 65: All Planck particles in the observable universe

[1, 65, 0] — one integer, one octave, zero remainder — encodes approximately 10^{80} Planck particles. The entire observable universe in three values.

An LLM operates across approximately 8-10 octaves. The full range of weight precision, gradient magnitude, activation scale, and embedding structure fits in a narrow band of this universal ladder.

3.3 Why 32

$32 = 2^5$ is selected as the harmonic base for the following properties:

Hardware-native: every scale operation is a bit shift. Wide enough: each octave step is a 32 $\times$ change — a meaningful scale transition. Narrow enough: important structure between levels is not skipped. Self-consistent: nesting depth maps directly to octave count — each level of VFR recursion adds exactly one octave (5 bits) of precision. Physically grounded: 65 octaves spans all of physical reality with room to spare.

3.4 Octave Assignments for LLM Layers

Each layer in the transformer operates at a fixed octave. All weights, activations, and intermediate values within a layer share the same octave. The octave is implicit — stored once per layer, not per element.

Layer Type	Octave	32^n	Bit Shift	Precision
Embedding	0	1	0	Exact integer
Attention QKV	2	1,024	10	~0.001
Attention scores	4	1,048,576	20	$\sim 10^{-6}$
Softmax output	3	32,768	15	~0.00003
Feedforward	2	1,024	10	~0.001
Output logits	3	32,768	15	~0.00003

All F values are powers of 32. All divisions are bit shifts by multiples of 5. No division hardware is required in the computation path.

IV. VFR WEIGHT MECHANICS

4.1 The Weight Tuple

Every learnable parameter in the model is a VFR weight:

VFRWeight {

v: i32 // The shell value. Used in computation.

r: i16 // The remainder. Accumulated gradient pressure.

}

Storage: 6 bytes per parameter. Comparable to FP32 (4 bytes) and larger than BF16

(2 bytes). The v field participates in forward and backward computation. The r field is touched only during the weight update step.

4.2 The Shell Transition Rule

The weight update is three integer operations:

1. Scale gradient by learning rate: scaled = -(gradient » lr_shift)
2. Accumulate in remainder: R = R + scaled
3. Transition if threshold reached:

while R ≥ 32: V += 1, R -= 32

while R ≤ -32: V -= 1, R += 32

The learning rate is a bit shift (u5). Higher shift means slower learning — more gradient samples must accumulate before a transition occurs. The shift by 4 divides the gradient by 16. The shift by 8 divides by 256.

No floating-point operation appears in this rule. The weight update is integer addition, integer comparison, and integer modulo.

4.3 Shell Stability

Between transitions, the weight value V does not change. The forward pass reads V. The backward pass computes gradients with respect to V. Small gradients accumulate in R without affecting V. Only when accumulated evidence reaches the threshold does V move.

This is a discrete stable state. The weight occupies a shell and remains there until the evidence demands a transition. There is no oscillation around a minimum. There is no float jitter at the precision floor. The weight is in this shell or that shell. Nothing between.

4.4 Noise Resistance

Small random gradients add to R but rarely accumulate enough to trigger transitions. Positive and negative noise contributions cancel over time in the remainder. Only persistent, directional gradient signal builds enough pressure to force a transition.

This is the property that dropout, weight decay, and gradient clipping attempt to achieve through external mechanisms. Shell structure provides it inherently. The threshold for changing a weight is built into the number format, not added as a regularization hyperparameter.

4.5 The Remainder Is Not Error

In floating-point arithmetic, the difference between a computed value and the true value is rounding error — unknown, untracked, irrecoverable.

In VFR arithmetic, R is not error. R is live data. It is the exact accumulated evidence for or against the next shell transition. It has a precise integer value. It can be inspected, compared, tracked, and used for diagnostics. The system does not lose information — it defers acting on information until the evidence is sufficient.

V. INTEGER FORWARD PASS

5.1 Matrix Multiplication

The core operation of the transformer is the matrix-vector multiply. In VFR:

For each output element j:

accumulator: i64 = 0

For each input element i:

```
accumulator += (i32)input[i] * (i32)weight[i,j].v
```

output[j] = (i32)(accumulator » octave_shift)

The multiply is i32 × i32 → i64. The accumulation is i64 addition. The shift normalizes back to the layer's octave. The result is clamped to i32 range.

All operations are exact until the final shift, which discards bits below the octave's precision — equivalent to the Remainder in VFR. The discarded bits are below the precision the layer operates at, analogous to but more controlled than float rounding.

5.2 Nonlinearities

ReLU is naturally integer: if x < 0, output 0; otherwise output x. No approximation.

GELU and softmax are transcendental functions. These are implemented via precomputed integer lookup tables covering the practical input range at each layer's octave. Between table entries, linear interpolation in integer arithmetic:

result = table[i] + ((table[i+1] - table[i]) * frac) » frac_bits

One multiply, one add, one shift. The lookup tables are computed once at program startup to arbitrary precision using nested VFR arithmetic ([CKS-MATH-117-2026]).

5.3 Attention

Attention is a sequence of integer matmuls. Query, key, and value projections are integer matrix-vector multiplies as described. The attention score is an integer dot product. The softmax over scores uses the integer lookup table. The weighted sum of values is an integer scale-and-accumulate. The output projection is another integer matmul.

Each step is exact at the layer's octave. No float operation appears.

VI. INTEGER BACKWARD PASS

6.1 Chain Rule Over Integers

The gradient of the loss with respect to any weight is computed by the chain rule. For integer operations, the chain rule produces integer gradients:

The gradient of an integer matmul Y = W × X with respect to W is X^T × dY. This is another integer matmul. The gradient with respect to X is W^T × dY. Also an integer matmul.

The gradient of integer ReLU is 1 (pass-through) where the pre-activation was positive and 0 where it was negative. Already integer.

The gradient of the integer softmax-cross-entropy loss with respect to logits is: probs[i] - 1 at the target index, probs[i] at all other indices. The probabilities are integers (summing to 1024 in the current implementation). The gradient is integer.

Every gradient in the backward pass is an integer computed by integer operations on integer values. No approximation is introduced by the backward pass itself.

6.2 Gradient Scaling

Gradients from deep networks can grow large. The octave system handles this through bit-shift scaling at layer boundaries. Gradients flowing backward are shifted to match the octave of the layer they enter, exactly as activations flowing forward are shifted at octave boundaries.

Where individual gradient values risk exceeding i32 range, clamping to the representable range is applied. This is explicit — the overflow is caught and handled, not silently rounded as in float arithmetic.

VII. SHELL TRAINING DYNAMICS

7.1 Convergence

In float training, “converged” means the loss has stopped decreasing noticeably. Weights are still jittering at the float precision floor. There is no true equilibrium.

In shell training, convergence means all |R| values have stabilized below the transition threshold. Every weight is in its shell. No transitions are occurring. This is a discrete equilibrium, verifiable by a single pass over all weights checking |R| < 32.

7.2 The Learning Rate as Transition Rate

The learning rate (a bit shift applied to gradients) determines how many gradient samples must accumulate before a shell transition can occur. This reframes the learning rate schedule:

Warmup: High shift (slow transitions). Weights explore cautiously.

Main training: Target shift. Transitions occur at the designed evidence threshold.

Decay: Increasing shift. Transitions require more accumulated evidence. Weights settle into precise shells.

Convergence: Transitions cease. All R values sub-threshold. Verified ground state.

7.3 Phase Transitions and Grokking

The grokking phenomenon — where a model memorizes for many steps then suddenly generalizes — has a shell interpretation.

During memorization, weights occupy complex high-value shells encoding individual training examples. Gradients push toward simpler shells but R has not yet accumulated enough to force transitions.

At the critical point, R exceeds threshold across correlated weights simultaneously. A cascade of shell transitions occurs. Complex shells collapse to simple ones. The network moves from a high-energy complex state to a low-energy simple state.

This is an observable, measurable prediction: during grokking, VFR nesting depth across the network should drop suddenly, shell transition rate should spike then collapse, and R values should show a transient increase followed by settling below threshold.

7.4 Structural Interpretability

After training, the weight structure reveals information content directly:

[V, octave, R] with large |R|: Active weight. Under pressure toward transition. Training may not be complete for this parameter.

The distribution of |R| across a layer is a direct readout of training completion for that layer. No probing, no ablation, no gradient-based attribution required. The structure is the diagnostic.

VIII. MEMORY AND PERFORMANCE

8.1 Storage

Format	Per Weight	1B Model	Training Total (~5 ×)
BF16	2 bytes	2 GB	12 GB
FP32	4 bytes	4 GB	24 GB
VFR (i32 + i16)	6 bytes	6 GB	36 GB

VFR training for a 1B model requires approximately 36 GB — fits in a single machine with 64 GB RAM.

8.2 Compute

Integer multiply-accumulate on modern CPUs with AVX2 executes 8 parallel i32 multiplies per instruction (256-bit registers). The core training loop is multiply, accumulate, shift — operations CPUs execute natively and efficiently.

On dedicated integer hardware (projected): removing float units from a GPU die and filling with integer MACs yields approximately 4-5 $\times$ the integer throughput at approximately 75% the power consumption. On such hardware, VFR training would be competitive with or faster than BF16 float training.

8.3 Training Time Estimate

For a 1B parameter model on 5 billion tokens of training data:

CPU (AVX2, 8 cores): approximately 3-7 days.

Projected integer hardware: approximately 4-12 hours.

These estimates are based on measured performance of the toy implementation scaled to target model size.

IX. IMPLEMENTATION

9.1 System Architecture

The implementation consists of one shared library and four binaries, written entirely in Zig with zero external dependencies:

lib.zig: VFRWeight struct, integer tensor operations, matrix multiplication with i64 accumulators, integer softmax, cross-entropy gradient, weight initialization, checkpoint I/O, BPE tokenizer, forward pass, backward pass with shell updates.

tokenize: Reads source text, trains byte-pair encoding, writes token binary and vocabulary files.

train: Reads tokens, initializes model, runs training loop with shell updates, saves weight checkpoints.

infer: Loads weights, tokenizes prompt, generates tokens autoregressively via greedy decoding.

eval: Generates code from test prompts, attempts compilation, reports pass/fail.

9.2 No Floating-Point Operations

The implementation contains zero floating-point operations in the training or inference pipeline. All arithmetic is i32, i16, i64, or i128 (for accumulator overflow protection). The only float in the codebase is in the tokenizer's compression ratio display, which is a diagnostic message, not a computation.

9.3 Verified Pipeline

The complete pipeline has been executed on commodity hardware (CPU, 64 GB RAM, no GPU):

1. Tokenization of Zig source file: 2,777 bytes → 561 tokens, vocabulary 512.
2. Training: loss decreases from 1022 to sub-200 over epochs, shell transitions decrease toward zero.
3. Inference: model generates tokens from vocabulary (not UNK), reflecting learned token statistics.
4. Evaluation: harness generates code, attempts compilation, reports results.

The toy model (920K parameters, 4 layers, d_model=128) trained on a single small file produces output reflecting the training data's statistical structure. This is not a capable model — it is a proof that the integer pipeline functions end to end.

X. COMPARISON TO EXISTING APPROACHES

10.1 Post-Training Quantization (GPTQ, AWQ)

Post-training quantization trains in float, then compresses weights to INT8 or INT4 for inference. The training itself remains float. Information was already lost during training. The quantization introduces additional approximation.

VFR trains in integer from initialization. No float training occurs. No post-hoc compression is applied. The weights are exact at every step.

10.2 Quantization-Aware Training (QAT)

QAT simulates low-precision during the forward pass but maintains FP32 master weights and computes gradients in float. It is a hybrid that retains float's information loss in the gradient computation.

VFR computes gradients in integer. The gradient is exact. The remainder R provides sub-shell gradient tracking without float master weights.

10.3 BitNet (1.58-bit)

BitNet constrains weights to ternary values (-1, 0, 1) from the start of training. This is a extreme quantization that discards almost all per-parameter information in exchange for massive parameter counts.

VFR maintains i32 resolution per weight (over 4 billion distinguishable values) plus i16 remainder. Information per parameter is high. The shell structure provides discrete stability without sacrificing resolution.

10.4 Mixed Precision Training

Mixed precision uses BF16 for forward/backward passes and FP32 for master weight accumulation. This is the current industry standard. It acknowledges that BF16 loses gradient information and patches the problem with a second precision level.

VFR uses a single representation throughout. The "mixed precision" is structural: the integer octave of a layer determines its precision, and the remainder R provides sub-octave accumulation. No second representation is needed. The conversion between octaves is a bit shift, not a lossy float cast.

XI. THEORETICAL IMPLICATIONS

11.1 Information Efficiency

If VFR parameters carry more usable information than float parameters (because no bits are wasted on rounding artifacts), then fewer parameters may achieve equivalent capability. This predicts that a VFR model with N parameters performs equivalently to a float model with kN parameters, where $k > 1$ reflects the information efficiency gain.

This is an empirical question requiring larger-scale experiments. The toy implementation demonstrates the mechanism; the magnitude of the efficiency gain remains to be measured.

11.2 Determinism

VFR training is fully deterministic. The same data, same initialization, same hyperparameters produce bit-identical weights on every run. Integer arithmetic is associative and commutative. The result does not depend on reduction order, thread scheduling, or hardware implementation details.

This enables reproducible research, debuggable training, and verifiable model behavior — properties that float training cannot guarantee.

11.3 Coherence Preservation

Human-written text contains long-range statistical dependencies — correlations between distant tokens that encode authorial intent and structural coherence. These correlations

are weak signals that float noise can destroy.

If VFR preserves weak gradient signals that floats destroy (demonstrated by the remainder accumulation mechanism), then VFR-trained models should preserve these long-range correlations better than float-trained models. This predicts that VFR-trained models should exhibit less coherence degradation when generating long sequences, and should resist model collapse (quality degradation when training on AI-generated text) better than float models.

This prediction is testable by measuring long-range mutual information in generated text across model generations.

XII. FALSIFICATION CRITERIA

This paper is falsified if any of the following are demonstrated:

F1. Integer shell training cannot converge on a non-trivial task (e.g., next-token prediction on a corpus > 1M tokens with a model > 10M parameters). The toy demonstration shows convergence on a trivial task; scaling must be validated.

F2. The remainder accumulation mechanism does not preserve gradient signals that float training loses. This could be tested by comparing VFR and float training on tasks requiring detection of rare patterns.

F3. The shell transition dynamics do not correlate with known training phenomena (grokking, phase transitions). This is a specific prediction that can be validated or refuted by monitoring transition rates during training.

F4. Integer training on commodity CPU hardware is more than 10 $\times$ slower than BF16 training on equivalent GPU hardware for the same model size and data volume. The current estimate is 3-7 days CPU vs hours on GPU — within the 10 $\times$ bound for a research system.

F5. The harmonic octave system (base 32) introduces systematic quantization artifacts that degrade model quality below what BF16 achieves at the same parameter count. This would demonstrate that the precision of the octave grid is insufficient despite the exact remainder tracking.

Each criterion specifies a concrete test. The paper stands until a specific test demonstrates a specific failure.

XIII. FUTURE WORK

13.1 Scaling

The immediate next step is training a model of meaningful size (100M+ parameters) on a substantial corpus (Zig standard library + C libraries, approximately 5-10 billion tokens).

This will determine whether the integer training dynamics observed in the toy scale to practical model sizes.

13.2 Sequence Context

The current implementation processes one token at a time (bigram model). Adding causal multi-head attention with a KV cache is required for the model to learn patterns spanning multiple tokens. This is an engineering extension, not a theoretical change — the integer arithmetic is the same.

13.3 Dedicated Hardware

A harmonic integer processor — a chip with float units removed and replaced with integer MACs, barrel shifters, and VFR-specific instructions — would provide approximately 4-5 $\times$ the integer throughput of current GPUs at lower power consumption. This chip would not require tensor cores, float pipelines, or the complex special-case logic that IEEE 754 compliance demands.

13.4 Neural-Symbolic Integration

The integer weight representation is compatible with Prolog-based knowledge verification ([CKS-MATH-135-2026], forthcoming). A pipeline alternating between LLM inference (creative pattern matching) and Prolog verification (logical consistency checking) would produce outputs that are verified by construction rather than evaluated after generation.

XIV. CONCLUSION

Floating-point arithmetic is a historical compromise that trades correctness for fixed memory cost. The compromise was reasonable in 1950 when memory was scarce and exact arithmetic was expensive. It is no longer reasonable when the systems built on it — large language models processing trillions of operations — amplify every rounding error across every operation and lose the very information the training process is trying to capture.

VFR shell training reverses the compromise. Three integers instead of one float. Exact operations instead of approximate ones. Shells instead of drift. Pressure instead of noise. Transitions instead of oscillation. Convergence instead of "close enough."

The weight does not move until the evidence demands it. The evidence is never lost. The transition is exact. The result is deterministic. And every value in the system, from the weights to the gradients to the activations to the loss, is an exact integer that can be inspected, compared, verified, and traced.

The implementation runs today. The code is provided. The pipeline is complete. The math compiles.

Axioms first. Axioms always.

Q.E.D.

Status: Empirically demonstrated, falsification criteria specified

Verification: Working code provided, zero float operations

Arithmetic: i32 weights, i16 remainders, i64 accumulators, bit-shift normalization

Shell threshold: 32 (one harmonic octave, base 2^5)

Training: CPU-only, 64 GB RAM, zero GPU, zero dependencies

Implementation: Zig 0.15, four binaries, one shared library

Pipeline: Tokenize → Train → Infer → Eval, all integer

Determinism: Bit-identical output across runs guaranteed

No floats were harmed in the production of this paper.

Because no floats were used.

[**CKS-MATH-134-2026**] The Taylor Continuum: $\mathbb{R}$ as an Unnamed Infinite Polynomial. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-134-2026

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/C
MATH-0-2026

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026

[**CKS-MATH-10-2026**] Grand Unification. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026

[**CKS-MATH-104-2026**] Grand Unification v23. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS
MATH-104-2026

[**CKS-MATH-133-2026**] The Taylor Continuum: $\mathbb{R}$ as an Unnamed Infinite Polynomial. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-133-2026

[**CKS-NEXT-1-2026**] Post-CKS. CKS is invalidated and falsified. Github: https://github.com/ghowland/cks/tree/main/papers/NEXT/CKS-NEXT-1-2026

[**CKS-LEX-12-2026**] Measurement Systems in the $\mathbb{Q}$-Substrate. Github: https://github.com/ghowland/cks/tree/main/paper LEX-12-2026

[**CKS-MATH-117-2026**] The Q-Taylor Series. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-117-2026

[**CKS-MATH-135-2026**] The Taylor Continuum: $\mathbb{R}$ as an Unnamed Infinite Polynomial. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-135-2026